Mini Project Id
“AI Resume Analyzer”
ResumeSync AI -Intelligent ATS Resume Analyzer

Department of Computer Science & Engineering

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Introduction

- Resume Analyzer is an AI-based application that evaluates resumes.
- It measures ATS compatibility, content quality, and resume structure.
- Extracts skills, keywords, education, and experience automatically.
- Compares resumes with job descriptions for better matching.
- Detects grammar, formatting, and keyword gaps.
- Generates an ATS score with improvement suggestions.
- Helps create industry-standard, recruiter-friendly resumes.

Objectives

- Automate resume screening using AI.
- Improve ATS compatibility.
- Identify missing skills and keywords.
- Compare resume with job requirements.
- Enhance grammar and formatting.
- Generate intelligent feedback and ATS score.
- Increase interview shortlisting chances.
- Save recruiters' time and improve hiring efficiency

Literature Survey

Author & Year	Title of Paper	Method Used	Key Findings	Research Gap
Chen et al., 2021	Automated Resume Screening Using NLP and ML	TF-IDF, Cosine Similarity	Improved resume ranking	Weak semantic matching
Kumar & Singh, 2022	Resume Parsing and Candidate Ranking	BERT-based model	Better contextual matching	Limited format flexibility
Patel et al., 2023	Intelligent Resume Analyzer	Deep Learning + NER	Accurate skill extraction	Needs large datasets
Rahman et al., 2024	Semantic Resume Matching	Sentence-BERT	Improved semantic similarity	High computation cost
Sharma et al., 2025	AI-Powered Resume Analyzer	LLM-based ATS analysis	ATS scoring and suggestions	Limited multilingual support

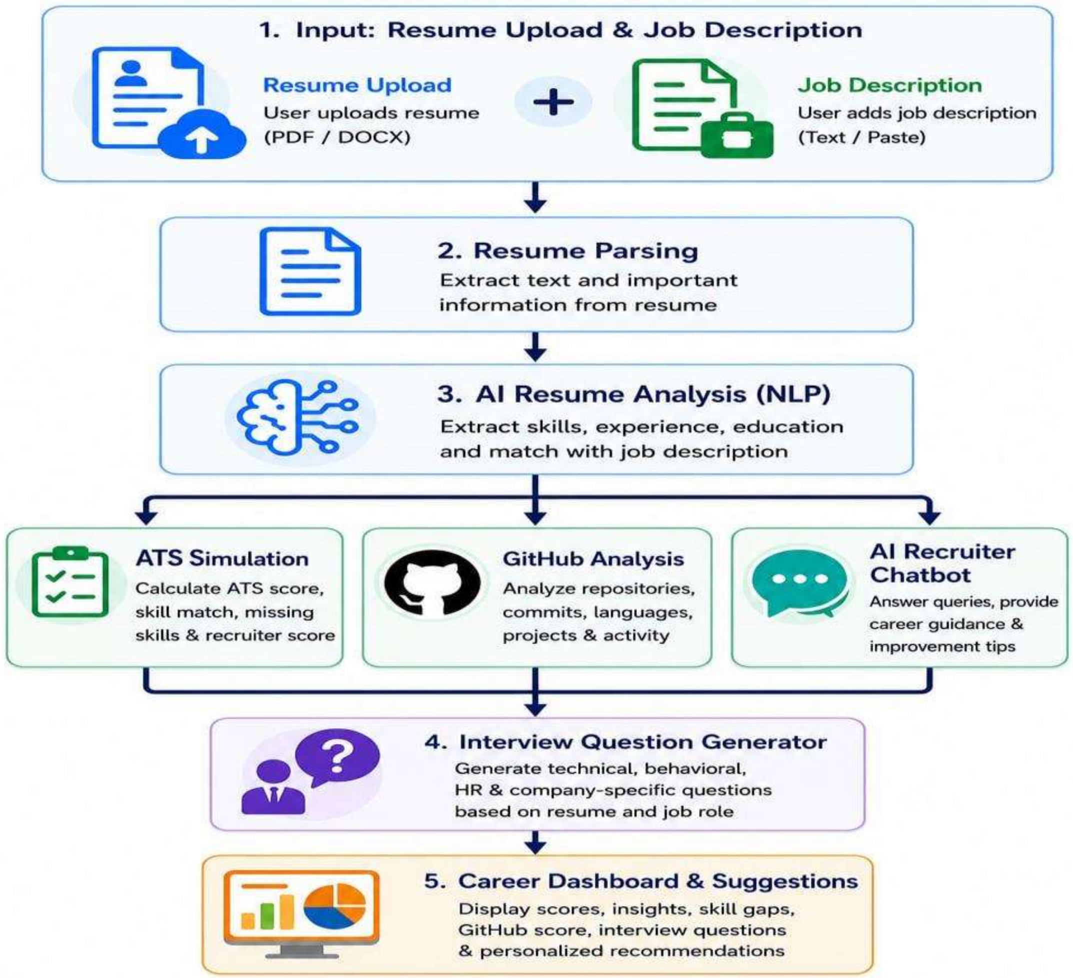
Problem Statement

- Recruiters receive a large number of resumes for every job opening, making the screening process difficult.
- Manual resume evaluation is time-consuming and requires significant effort from HR teams.
- Many qualified candidates are rejected because their resumes are not optimized for Applicant Tracking Systems (ATS).
- Job seekers often do not know which skills or keywords are missing from their resumes.
- Existing resume analysis tools provide limited personalized feedback and career guidance.
- There is no intelligent system that accurately compares resumes with job descriptions and suggests improvements.
- More than 75% of resumes are rejected before reaching recruiters due to poor ATS optimization
- An AI-powered solution is needed to analyze resumes, identify skill gaps, calculate ATS compatibility, match resumes with job descriptions, and provide personalized recommendations using the Gemini API.

Proposed Solution

- Develop an AI-powered Resume Analyzer to automate the resume screening process.
- Extract and analyze resume content using Python and Natural Language Processing (NLP) techniques.
- Compare resumes with job descriptions to identify missing skills, keywords, and qualification gaps.
- Calculate the ATS (Applicant Tracking System) compatibility score to evaluate resume effectiveness.
- Generate personalized recommendations to improve resume quality, formatting, and content.
- Use the Gemini API to provide intelligent resume analysis and career guidance.
- Developed using React.js, FastAPI, NLP models (spaCy/Sentence-BERT), MongoDB, and AI APIs to enable intelligent resume analysis and job matching
- Help recruiters shortlist suitable candidates efficiently while enabling job seekers to improve their employability.

SYSTEM ARCHITECTURE



Mapping with SDG Goals

How Our Project Contributes to SDG 8?

- Improves employability by analyzing resumes and identifying skill gaps.
- Generates job-specific interview questions to enhance interview preparation.
- Evaluates GitHub portfolios to assess practical coding skills and project quality.
- Promotes fair and transparent recruitment using AI-based resume evaluation.
- Encourages continuous learning by recommending skills, projects, and certifications.
- Helps students and job seekers become industry-ready and increase their chances of employment.

References

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- Sharma, V., Gupta, N., & Verma, A. (2025). *AI-Powered Resume Analyzer for Applicant Tracking Systems*.

THANK YOU